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Unraveling Neural Dynamics of Syntactic Processing in First and Second Language Planning of Sentence Production

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Introduction

- During **sentence production**, preverbal messages are proposed to be **syntactically encoded** to construct well-organized structures before articulation, which is at the core of the intermediate formulating phase.
- Existing studies, however, have a rather circumscribed scope **focusing on spatial localization**. As a result, they can **scarcely elucidate the temporal dynamics** involved in incrementally generating complex syntactic structures for subsequent externalization.
- Recent advances in **electrophysiological methodologies** have begun to illuminate these **temporal dynamics of syntactic processing** during sentence production. However, significant gaps remain in our understanding of **neural dynamics underlying both L1 and L2 syntactic processing before articulation**.

Materials & Methods

Participants >>>

Right-handed native speakers of Mandarin Chinese (n = 33, 17 males, 21.24 ± 2.05 years) and Mandarin Chinese L2 learners with Korean as their native language (n = 32, 16 males, 24.06 ± 2.38 years) participated in this study.

Materials >>>

Subjective-relative clause (SRC)

[CP...三角[VP照了[NP[CP[IP(ti)[VP撞了椭圆]]的]方块i]]]。

T1on T4off T6off

Literal glosses: Triangle illuminate le (ti) hit le ellipse de square。

Translation: The triangle illuminates the square which hits the ellipse

Objective-relative clause (ORC)

[CP...三角[VP照了[NP[CP[IP椭圆[VP撞了(ti)]]的]方块i]]]。

T1on T4off T6off

Literal glosses: Triangle illuminate le ellipse hit le (ti) de square。

Translation: The triangle illuminates the square which the ellipse hits.

Fig.1 Experimental materials (examples)

Procedure >>>

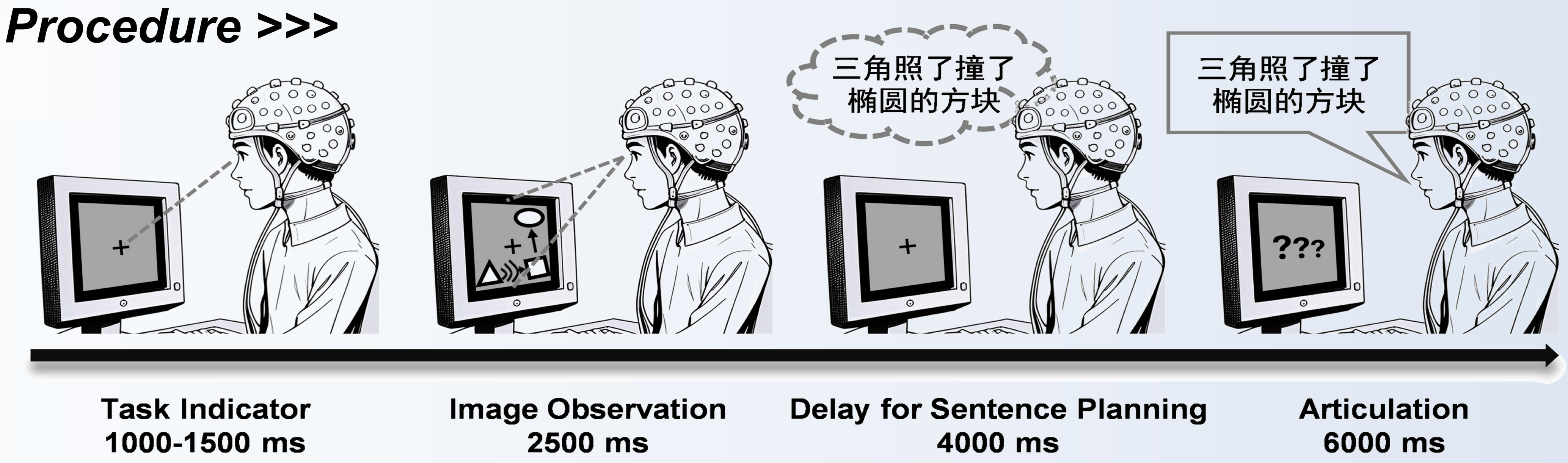

Fig.2 Experimental procedure with the specific timing parameters

- Participants were asked to produce **syntactically complex Mandarin Chinese sentences** containing SRC and ORC, both of which involve reordering and integration across long-distance filler-gap dependencies (**Fig.1**).
- Through a picture description paradigm (**Fig.2**) employing **abstract geometric objects and action symbols**, we characterized the neural dynamics of syntactic processing while **minimizing semantic interference**.

Results

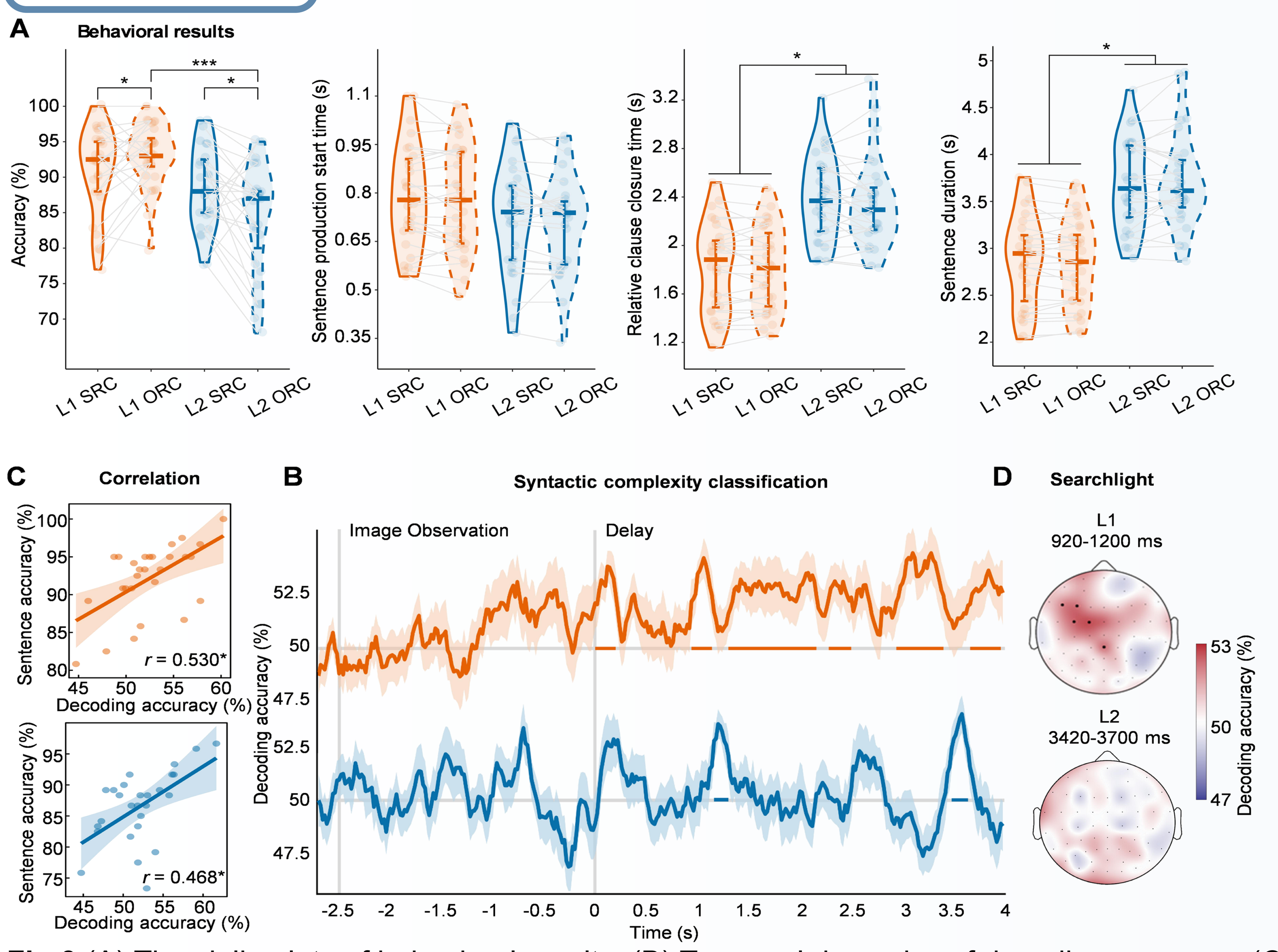

Fig.3 (A) The violin plots of behavioral results. (B) Temporal dynamics of decoding accuracy. (C) Correlations between decoding accuracy (L1: 920-1200 ms; L2: 3420-3700 ms) and average SRC and ORC accuracy for L1 and L2 participants. (D) Using the same time windows for searchlight analysis to reveal the electrodes' contributions. * $P < 0.05$; *** $P < 0.001$.

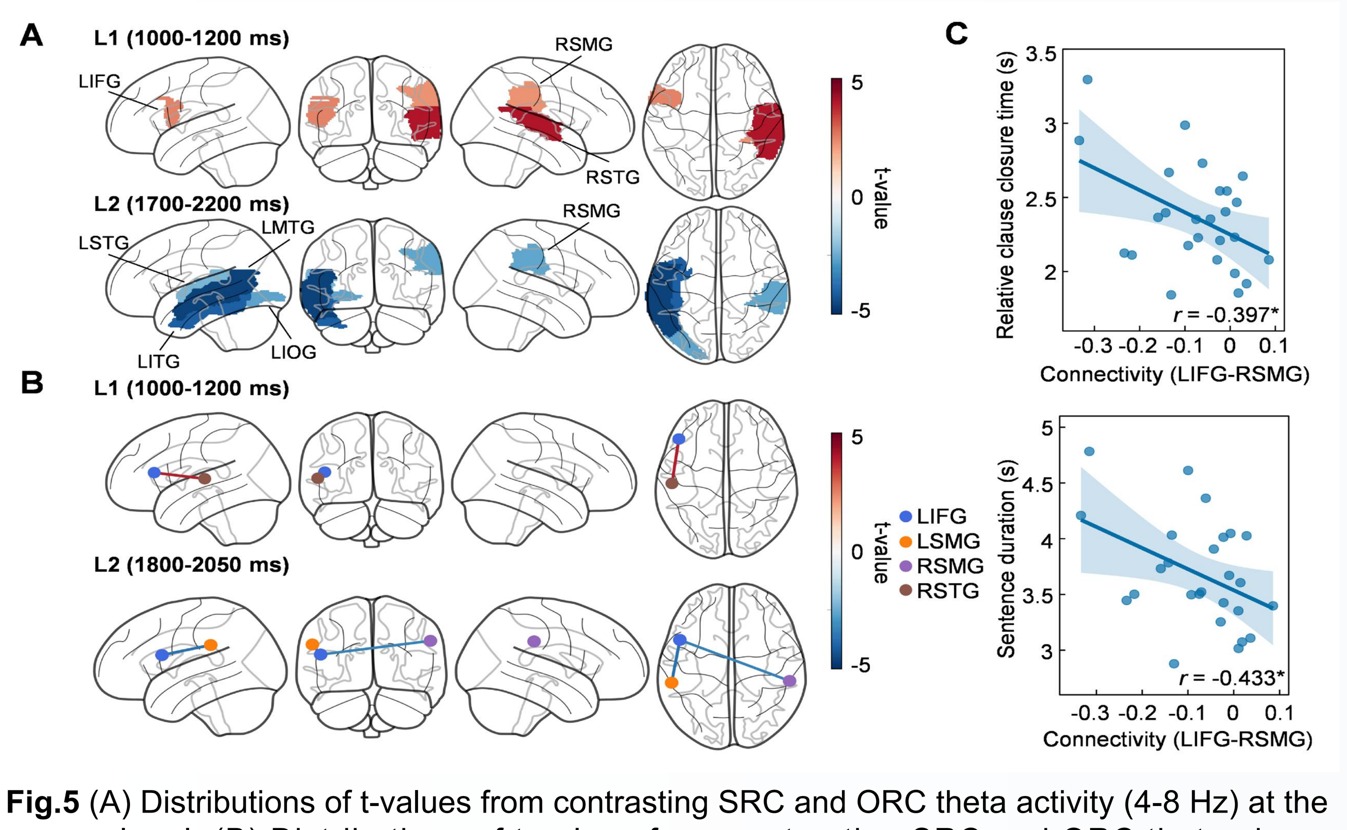

Fig.5 (A) Distributions of t-values from contrasting SRC and ORC theta activity (4-8 Hz) at the source level. (B) Distributions of t-values from contrasting SRC and ORC theta phase synchrony at the source level between LIFG and bilateral temporoparietal brain regions, masked at $P < 0.05$, FDR corrected. (C) Scatter plots illustrating the correlations between L2 participants' connectivity and RTs. * $P < 0.05$.

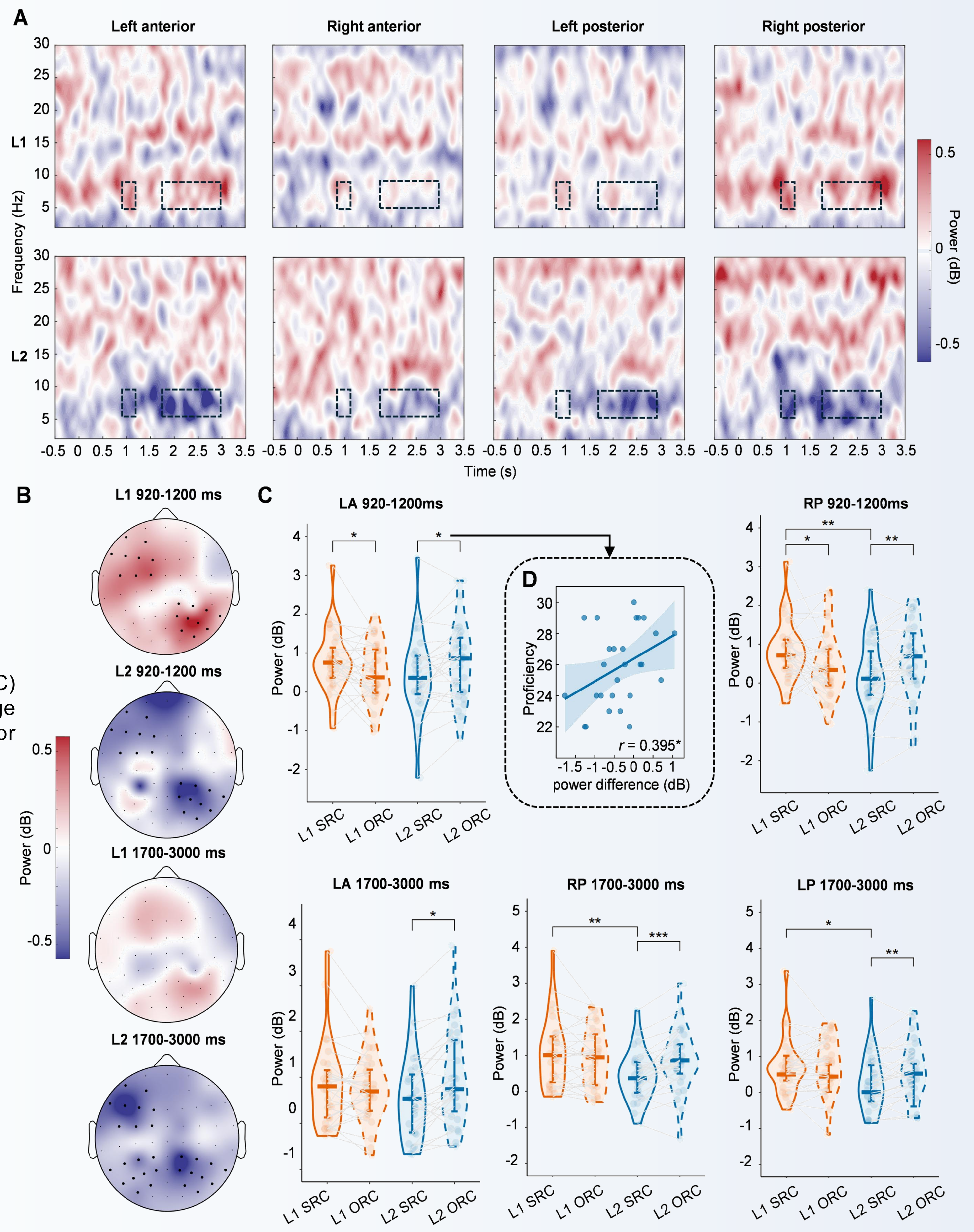

Fig.4 (A) Time–frequency representations (SRC minus ORC) for L1 and L2 participants. Black dashed squares represent the duration (920-1200 ms, 1700-3000 ms) used to calculate the averaged theta power. (B) Topographic maps of theta power differences (SRC minus ORC). Black dot clusters mark the ROI from Fig. 1D that reached statistical significance in ANOVA. (C) Violin plots showing averaged theta power in the SRC and ORC conditions across different ROIs and time windows. (D) For L2 participants, the theta power difference (SRC minus ORC) in LA (920-1200 ms) was significantly correlated with language proficiency. * $P < .05$; ** $P < 0.01$; *** $P < 0.001$.

Conclusion

1. L1 and L2 speakers exhibit fundamentally **different temporal dynamics** in syntactic processing during planning phase of sentence production.
2. **Theta oscillations** as the neural signature of syntactic computation during sentence planning, supporting their role in building complex syntactic structures.
3. L1 syntactic processing relies on specialized **left frontotemporal networks** during stence planning of production, while L2 syntactic processing recruits **broader bilateral networks**.

Key References

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